

CODEBOOK FOR QUALITATIVE ANALYSIS

Psychological Safety and Interactional Behaviors in Pair Programming

Supplementary research artifact for SBQS submission and Zenodo deposit

Artifact type	Qualitative analysis codebook
Version	1.0
Language	English with Portuguese code labels
Related manuscript	Psychological Safety in Pair Programming / Segurança Psicológica na Programação em Pares

1. Purpose and scope

This codebook documents the qualitative coding scheme used to analyze interpersonal risk-taking behaviors in industrial pair-programming transcripts. It is intended to support transparency, replicability, reviewer assessment, and reuse of the analytical procedure associated with the SBQS manuscript.

The codebook specifies the unit of analysis, episode-identification convention, coding dimensions, operational definitions, inclusion and exclusion rules, subtypes, decision rules, and a worked example drawn from the analysis matrix.

2. Data source and unit of analysis

The analysis uses publicly available transcripts from the PP-ind repository. The unit of analysis is an interaction episode: a continuous conversational segment focused on the same technical problem, decision, or topic and containing at least one interpersonal risk-taking manifestation.

Episodes begin when a new problem, doubt, or clear topic shift is introduced. They end when the issue is resolved, abandoned, or replaced by a definitive change of focus.

3. Episode identifier convention

Each episode identifier follows the pattern COMPANY–SESSION–EPISODE. In the analysis matrix this pattern is compacted as <Company><Session>-E<order>.

Component	Meaning	Example
Company letter	Company represented in the PP-ind dataset	A, B, C, D, E, J, O, or P
Session code	Transcript/session selected within the company	A1, A2, or A3
E	Interaction episode marker	E
Episode number	Sequential order of the episode in the transcript	01, 02, 03, ...
Line range	Start and end lines in the source transcript	L313-L320

Example: DA2-E16 identifies Company D, session/transcript A2, episode 16. The associated range L313-L320 indicates the source-transcript lines containing that episode.

4. Analytical dimensions

Each coded occurrence is represented by four linked dimensions: (1) interpersonal risk-taking behavior (RQ1), (2) partner reaction, (3) impact on task progression (RQ2), and (4) psychological-safety indicator.

4.1 Interpersonal risk-taking behaviors (RQ1)

Portuguese label	English label	Operational definition	Decision rule	Example
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Fazer perguntas	Asking questions	Explicit request for clarification, guidance, confirmation, or explanation regarding the task, the code, the domain, or a technical decision.	Code when the question reveals a gap in understanding or requests support from the partner. Merely rhetorical questions should not be coded.	"What is an anonymous class?"
Expressar incerteza	Expressing uncertainty	Expression of doubt, lack of knowledge, hesitation, or low confidence regarding one's own understanding, the code, or the proposed solution.	Includes admissions of lack of knowledge and expressions such as "I think," "I don't know," "maybe," or "I might be wrong," when they convey relevant vulnerability.	"Of course, I might be wrong too."
Levantar preocupações	Raising concerns	Identification of a risk, limitation, adverse effect, cost, insufficient performance, or possible negative consequence of the solution.	Code when the main function of the statement is to alert the partner to a potential problem, even when an alternative is also mentioned.	"The Contents are not there. That is bad."
Apontar erros	Pointing out errors	Correction, challenge, or indication of an error in the code, interpretation, decision, or reasoning presented.	The criticism must be directed at the action, artifact, or reasoning. Disagreement without an explicit indication of error should be classified as either Challenging or pointing out errors or Raising concerns, according to its predominant function.	"That is Swing." / "No, it is, in principle, AWT."
Propor mudanças	Proposing changes	Suggestion of an alternative, modification, refactoring, improvement, or new strategy for the solution or task execution.	Code when the statement introduces a concrete alternative course of action. An operational instruction should only be coded as Proposing changes when it alters the current strategy.	"Do it on localhost instead of dev-intern."

4.2 Partner reactions

Portuguese label	English label	Operational definition
Suporte/Validação	Support/Validation	The partner responds supportively to the manifestation by providing help, explanation, confirmation, agreement, constructive correction, or joint investigation.
Neutralidade	Neutrality	The partner provides only a minimal response, does not engage substantively, does not respond clearly, or shifts the focus without addressing the manifestation.
Rejeição	Rejection	The partner invalidates or interrupts the manifestation, criticizes the person, shows irritation directed at the interlocutor, or defensively closes the conversation.

4.3 Impact on task progression (RQ2)

Portuguese label	English label	Operational definition
Avançou	Advanced	The interaction contributed to task progression by resolving the problem, supporting a decision, enabling the task to continue, improving understanding, refining the solution, increasing its robustness, or supporting the evaluation of alternatives.
Travou	Blocked	The interaction hindered task progression by producing an impasse, loss of flow, interruption of the resolution process, or abandonment of the topic, problem, or line of reasoning without a clear resolution

4.4 Psychological-safety indicator

Portuguese label	English label	Operational definition
Sinal positivo	Positive signal	The interpersonal risk was received supportively, and the interaction fostered or preserved collaboration.
Sinal negativo	Negative signal	
Sinal ambíguo	Ambiguous signal	The manifestation received a defensive, hostile, or invalidating response, undermining collaboration.

5. Worked coding example

Episode DA2-E16 (Company D, session A2, episode 16; lines L313-L320) illustrates the full analytical path.

D3: “In an anonymous class? What is an anonymous class?”

D4: “Well, as soon as you implement, for example, a SelectionAdapter, that is an anonymous class because the class has no name.”

D3: “Oh, yes, right.”

Analytical element	Assigned label/evidence	Rationale
Source identifier	DA2-E16; L313-L320	Ensures traceability to the company, transcript, interaction episode, and corresponding transcript lines.
Initial behavior	Expressing uncertainty	D3 explicitly reveals a lack of knowledge about a technical concept.
Partner reaction	Support	D4 directly explains the concept and connects it to the code structure.
Task impact	Advanced	The response produces explicit learning, refines the participant’s understanding, and allows the task to continue.
PS indicator	Positive	The interpersonal risk is received supportively and leads to constructive collaboration.

6. Recommended matrix structure

Field	Recommended content
episode_id	Compact episode identifier, e.g., DA2-E16.
line_range	line_range
risk_behavior	risk_behavior
trigger_quote	trigger_quote
partner_response_quote	partner_response_quote
reaction_code	partner_reaction
impact_code	task_impact
ps_indicator	ps_indicator

7. Version history

Version	Date	Description
1.0	2026	Initial public version prepared as supplementary artifact for the SBQS submission.